

JOB PORTAL MANAGEMENT SYSTEM

Java + Hibernate + MySQL | Console-Based Application

Project Presentation

Includes Smart Skill-Based Job Recommendation

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Project Overview

- A console-based Job Portal application developed using Java, Hibernate ORM and MySQL.
- Provides CRUD operations for users, job posts and job technologies.
- Stores user skills and compares them with technologies required by jobs.
- The Smart Job Recommendation module calculates a job match percentage.
- Jobs are displayed in descending order of match percentage.

Objectives

- Manage users and their basic account information.
- Create, view, update and delete job postings.
- Manage technologies associated with each job post.
- Store skills possessed by individual users.
- Recommend jobs based on matching user skills with job requirements.
- Apply Hibernate ORM for database interaction instead of direct JDBC operations.

Technology Stack

- Programming Language: Java
- ORM Framework: Hibernate ORM 6.6.58.Final
- Database: MySQL 8.0.42
- Persistence API: Jakarta Persistence (JPA)
- Build/Dependency Management: Maven
- IDE: IntelliJ IDEA
- JDK: Java 21

Main Modules

1. User Management

Add User, View All Users, Search by ID, Update User, Delete User

2. Job Post Management

Add, View, Search, Update and Delete Job Posts

3. Technology Management

Add, View, View by Job Post ID, Update and Delete Technologies

4. Smart Job Recommendation

Add User Skills and View Recommended Jobs

Database Design

- users — stores user ID, username and password.
- job_post — stores job profile, description, required experience and user ID.
- job_tech_stack — stores technologies required for each job post.
- user_skill — stores skills associated with each user.
- The recommendation logic uses user_skill and job_tech_stack to compare skills.

Hibernate Architecture / DAO Layer

- Entity classes map Java objects to database tables using JPA annotations.
- DAO classes handle database operations through Hibernate Session and Transaction.
- HibernateUtil creates and provides the SessionFactory.
- Service layer contains recommendation/business logic.
- Main.java provides the console menu and connects user actions to the DAOs/services.

Smart Job Recommendation — Working

Step 1: Fetch all skills of the selected user.

Step 2: Fetch all available job posts.

Step 3: Fetch technologies required by each job.

Step 4: Compare user skills with required technologies.

Step 5: Calculate match percentage:

$$\text{Match \%} = (\text{Matched Skills} / \text{Required Technologies}) \times 100$$

Step 6: Sort jobs by match percentage in descending order.

Recommendation Example

User 1 skills: Java, Mysql

Job 1 technologies: Java, Hibernate, Mysql

Matched technologies: Java + Mysql = 2

Required technologies: 3

Match percentage = $(2 / 3) \times 100 = 66.67\%$

The same calculation is performed for every job that has technologies.